

The Randomized k -Server Conjecture is False!

Yuval Rabani - The Hebrew University of Jerusalem

Joint work with
Sébastien Bubeck (Microsoft Research) and Christian Coester (U. of Oxford)

Online computing (minimization problem)

- There's a two-person game between ADV and ALG, alternating moves.
- ADV pure strategies: sequences of moves, last move terminates the game. ADV has perfect recall, imperfect information (information sets group all ADV nodes reachable using same ADV strategy).
- ALG pure strategies: functions that map every alternating sequence of moves ending with an ADV non-terminating move to an ALG move. ALG has perfect information.
- At an end state: ALG cost c_{alg} and base (ADV) cost c_{base} . The base cost depends only on ADV information set.
- Typical base cost:

$$c_{\text{base}}(\sigma) = \inf_{\alpha} c_{\text{alg}}(\sigma, \alpha)$$

Competitive analysis

- The deterministic competitive ratio: the infimum over C s.t. $\exists$ pure ALG strategy α and a constant β for which $\forall$ pure ADV strategy σ ,

$$c_{\text{alg}}(\sigma, \alpha) \leq C \cdot c_{\text{base}}(\sigma) + \beta$$

- The randomized competitive ratio: the infimum over C s.t. $\exists$ mixed ALG strategy $\tilde{\alpha}$ and a constant β for which $\forall$ pure ADV strategy σ ,

$$E[c_{\text{alg}}(\sigma, \tilde{\alpha})] \leq C \cdot c_{\text{base}}(\sigma) + \beta$$

- The distributional lower bound: the supremum over C s.t. for every constant β , $\exists$ mixed ADV strategy $\tilde{\sigma}$ for which $\forall$ pure ALG strategy α ,

$$E[c_{\text{alg}}(\tilde{\sigma}, \alpha)] \geq C \cdot E[c_{\text{base}}(\tilde{\sigma})] + \beta$$

- Yao's minimax: randomized c.r. = distributional l.b.

Metrical task systems [Borodin, Linial & Saks 1987]

- Setting: n -point metric space $\mathcal{M} = (X, d)$, initial state $\alpha_0 \in X$
- Pure ADV strategies: sequences $\sigma_1, \sigma_2, \sigma_3, \dots$ where $\forall i$,
 $\sigma_i : X \rightarrow \mathbb{R}_+ \cup \{+\infty\}$
- Pure ALG strategies: sequences $\alpha_1, \alpha_2, \alpha_3, \dots$, where $\forall i$,
 $\alpha_i : \{\sigma_1, \sigma_2, \dots, \sigma_i, \alpha_1, \alpha_2, \dots, \alpha_{i-1}\} \rightarrow X$
- Cost:
$$c_{\text{alg}}(\sigma, \alpha) = \sum_{i \geq 1} (d(\alpha_{i-1}, \alpha_i) + \sigma_i(\alpha_i))$$
- Deterministic competitive ratio: $2n - 1$ [BLS87]
- Randomized competitive ratio: $O(\log^2 n) \cap \Omega(\log n / \log \log n)$
[BCLL19, BBM01, BLMN03]
Best existential l.b. (uniform metric spaces) $\Omega(\log n)$ [BLS87]

Lower bound: ρ -HSTs

- Definition: For $\rho \geq 1$, a metric space $\mathcal{M} = (X, d)$ is a ρ -HST iff there is a partition $X = X_1 \dot{\cup} X_2 \dot{\cup} \dots \dot{\cup} X_s$, such that $\exists \Delta > 0$ for which:
 - a. $\forall i \neq j$ and $x \in X_i$ and $y \in X_j$, $d(x, y) = \Delta$.
 - b. $\forall i$, $(X_i, d|_{X_i})$ is a ρ -HST with maximum distance $\leq \Delta/\rho$.
- $\mathcal{M}$ is a ρ -HST $\implies \mathcal{M}$ is an ultrametric $\implies \mathcal{M}$ is a 1-HST.
- Thm [BBM01]: The randomized competitive ratio for metrical task systems on every n -point $\Omega(\log^2 n)$ -HST is in $\Omega(\log n)$.
- Thm [BLMN03]: $\forall \epsilon \in (0, 1) \exists \delta > 0$ such that every n -point $\mathcal{M}$ contains an $n^{\delta/\log 2\rho}$ -point ρ -HST, up to bi-Lipschitz distortion $2 + \epsilon$.

Upper bound: 2-HSTs

- Thm [FRT03]: Every n -point metric space can be represented as a convex combination of dominating 2-HSTs, up to bi-Lipschitz distortion $O(\log n)$.
- Thm [BCLL19]: On every n -point 2-HST, the randomized competitive ratio for metrical task systems is in $O(\log n)$.

k -server problem [Manasse, McGeoch & Sleator 1988]

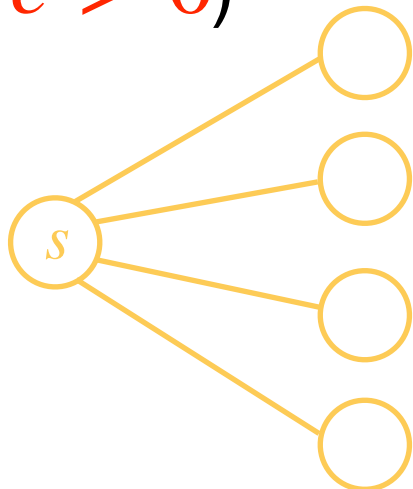
- Setting: metric space $\mathcal{M} = (X, d)$, initial configuration $\alpha_0 \in X^k$
 - Pure ADV strategies: sequences $\sigma_1, \sigma_2, \sigma_3, \dots$ where $\forall i, \sigma_i \in X$
 - Pure ALG strategies: sequences $\alpha_1, \alpha_2, \alpha_3, \dots$, where $\forall i, \alpha_i$ is a k -point configuration, and $\sigma_i \in \alpha_i$
 - Cost:
$$c_{\text{alg}}(\sigma, \alpha) = \sum_{i \geq 1} \min_{\pi: \alpha_{i-1} \leftrightarrow \alpha_i} \sum_{x \in \alpha_{i-1}} d(x, \pi(x))$$
 - Deterministic competitive ratio: between k [MMS88] and $2k - 1$ [KP95]
 - Randomized competitive ratio: $\Omega(\log k / \log \log k)$ [BBM01 + BLMN03]
- In $(k + 1)$ -point metric spaces: $O(\log^2 k)$ [BCLL19]
- In large metric spaces $\Omega(\log k)$ [KRR91, BKRS92, BBM01, BLMN03]

Layered graph traversal [Papadimitriou & Yannakakis 1989]

- Width- w layered graph, edge lengths $l(e)$, first layer $\{s\}$, last layer $\{t\}$
- When a layer is reached, the next layer, the edges between the layers, and their lengths are revealed
- Goal: starting at s , reach t ; minimize total length of walk
- Deterministically: $O(w2^w) \cap \Omega(2^w) \cdot d_l(s, t)$ [Bur96,FFKRRRV91]
- Probabilistically: $O(w^{13}) \cap \Omega(w^2/\log^{1+\epsilon} w) \cdot d_l(s, t)$ [Ram93]
(for all $\epsilon > 0$)

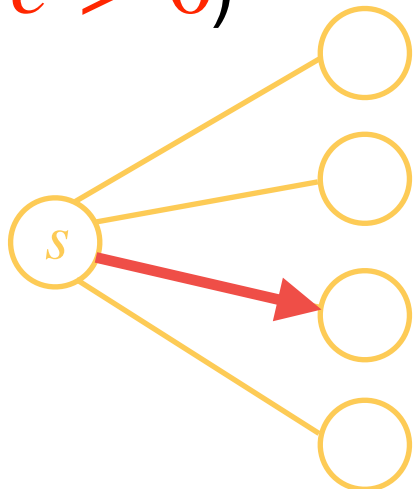
Layered graph traversal [Papadimitriou & Yannakakis 1989]

- Width- w layered graph, edge lengths $l(e)$, first layer $\{s\}$, last layer $\{t\}$
- When a layer is reached, the next layer, the edges between the layers, and their lengths are revealed
- Goal: starting at s , reach t ; minimize total length of walk
- Deterministically: $O(w2^w) \cap \Omega(2^w) \cdot d_l(s, t)$ [Bur96,FFKRRRV91]
- Probabilistically: $O(w^{13}) \cap \Omega(w^2/\log^{1+\epsilon} w) \cdot d_l(s, t)$ [Ram93]
(for all $\epsilon > 0$)



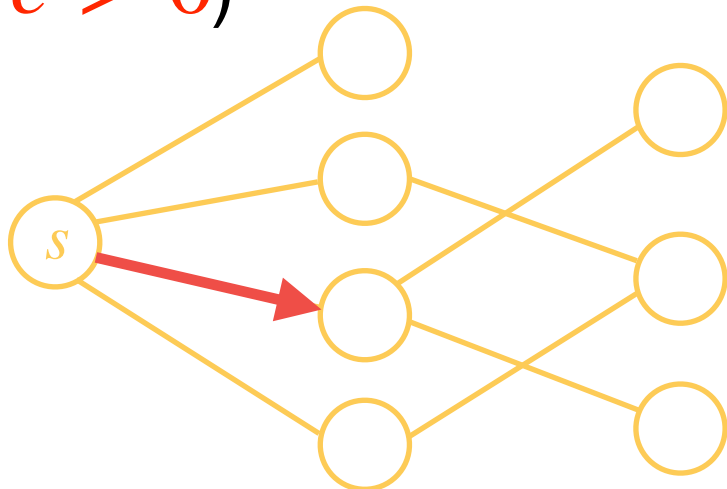
Layered graph traversal [Papadimitriou & Yannakakis 1989]

- Width- w layered graph, edge lengths $l(e)$, first layer $\{s\}$, last layer $\{t\}$
- When a layer is reached, the next layer, the edges between the layers, and their lengths are revealed
- Goal: starting at s , reach t ; minimize total length of walk
- Deterministically: $O(w2^w) \cap \Omega(2^w) \cdot d_l(s, t)$ [Bur96,FFKRRRV91]
- Probabilistically: $O(w^{13}) \cap \Omega(w^2/\log^{1+\epsilon} w) \cdot d_l(s, t)$ [Ram93]
(for all $\epsilon > 0$)



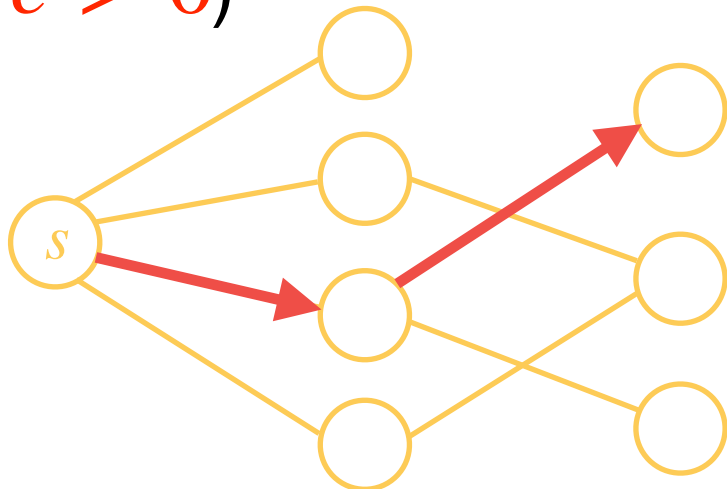
Layered graph traversal [Papadimitriou & Yannakakis 1989]

- Width- w layered graph, edge lengths $l(e)$, first layer $\{s\}$, last layer $\{t\}$
- When a layer is reached, the next layer, the edges between the layers, and their lengths are revealed
- Goal: starting at s , reach t ; minimize total length of walk
- Deterministically: $O(w2^w) \cap \Omega(2^w) \cdot d_l(s, t)$ [Bur96, FFKRRV91]
- Probabilistically: $O(w^{13}) \cap \Omega(w^2 / \log^{1+\epsilon} w) \cdot d_l(s, t)$ [Ram93]
(for all $\epsilon > 0$)



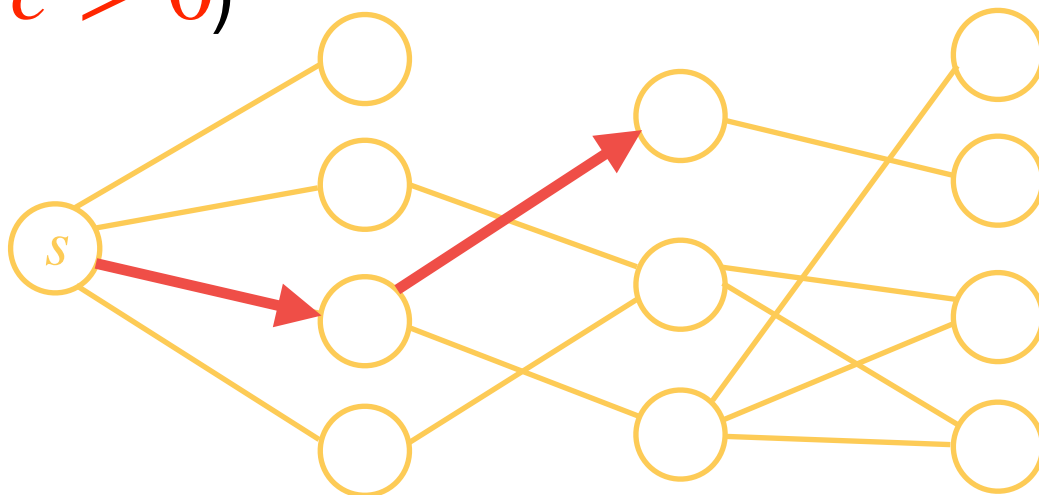
Layered graph traversal [Papadimitriou & Yannakakis 1989]

- Width- w layered graph, edge lengths $l(e)$, first layer $\{s\}$, last layer $\{t\}$
- When a layer is reached, the next layer, the edges between the layers, and their lengths are revealed
- Goal: starting at s , reach t ; minimize total length of walk
- Deterministically: $O(w2^w) \cap \Omega(2^w) \cdot d_l(s, t)$ [Bur96,FFKRRRV91]
- Probabilistically: $O(w^{13}) \cap \Omega(w^2/\log^{1+\epsilon} w) \cdot d_l(s, t)$ [Ram93]
(for all $\epsilon > 0$)



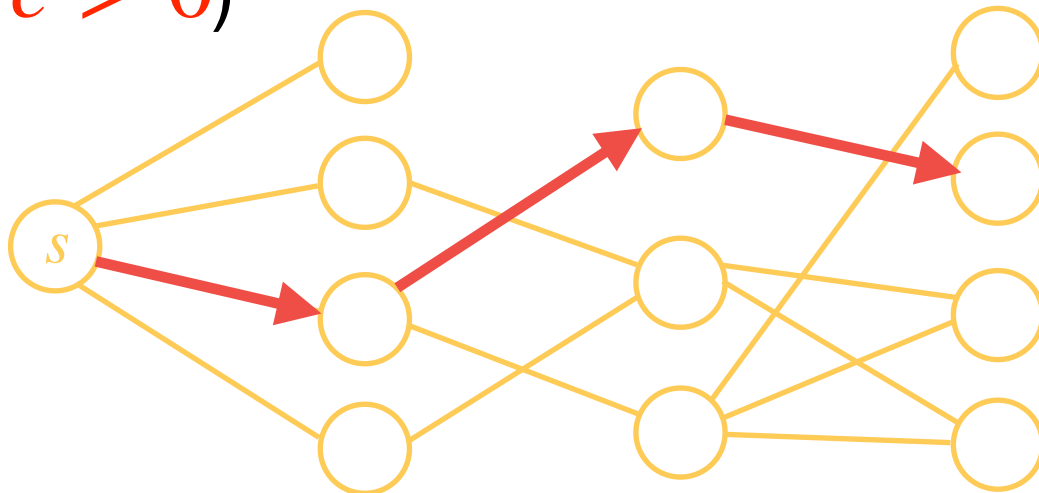
Layered graph traversal [Papadimitriou & Yannakakis 1989]

- Width- w layered graph, edge lengths $l(e)$, first layer $\{s\}$, last layer $\{t\}$
- When a layer is reached, the next layer, the edges between the layers, and their lengths are revealed
- Goal: starting at s , reach t ; minimize total length of walk
- Deterministically: $O(w2^w) \cap \Omega(2^w) \cdot d_l(s, t)$ [Bur96,FFKRRRV91]
- Probabilistically: $O(w^{13}) \cap \Omega(w^2/\log^{1+\epsilon} w) \cdot d_l(s, t)$ [Ram93]
(for all $\epsilon > 0$)



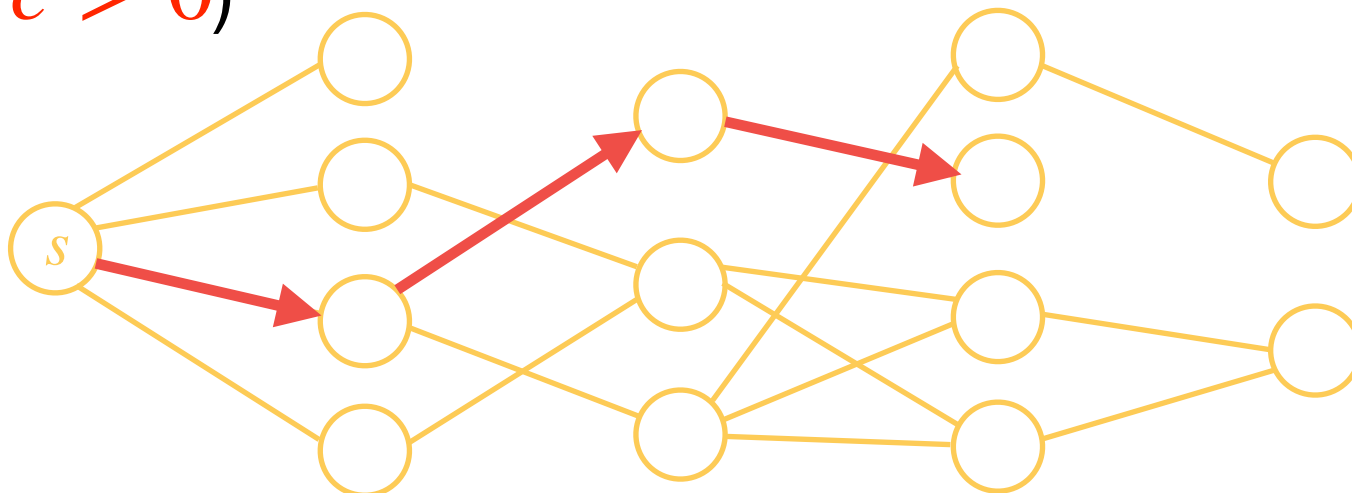
Layered graph traversal [Papadimitriou & Yannakakis 1989]

- Width- w layered graph, edge lengths $l(e)$, first layer $\{s\}$, last layer $\{t\}$
- When a layer is reached, the next layer, the edges between the layers, and their lengths are revealed
- Goal: starting at s , reach t ; minimize total length of walk
- Deterministically: $O(w2^w) \cap \Omega(2^w) \cdot d_l(s, t)$ [Bur96,FFKRRRV91]
- Probabilistically: $O(w^{13}) \cap \Omega(w^2/\log^{1+\epsilon} w) \cdot d_l(s, t)$ [Ram93]
(for all $\epsilon > 0$)



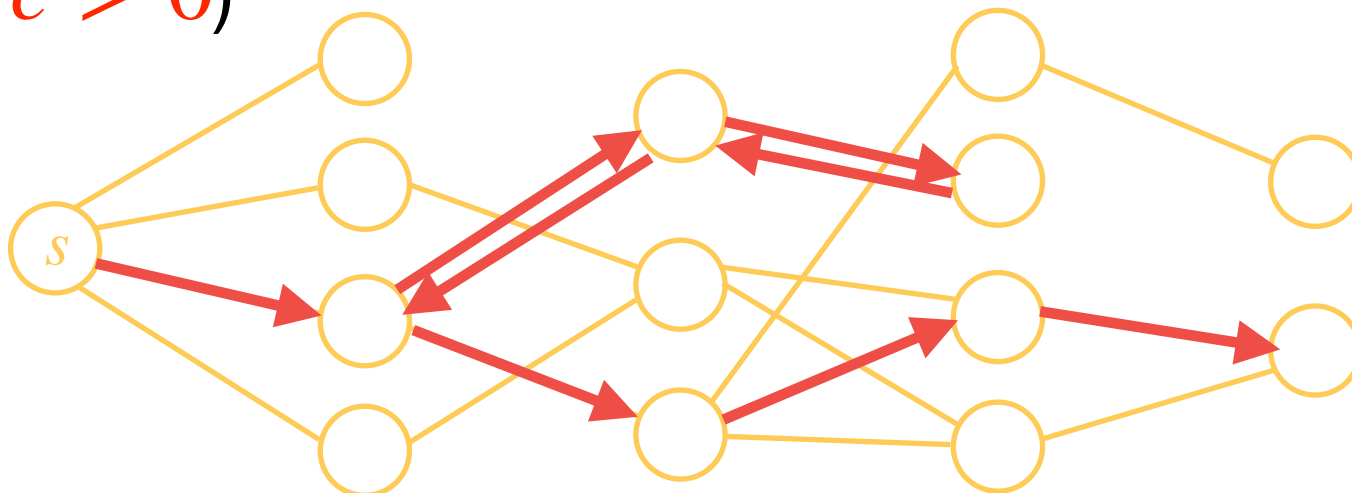
Layered graph traversal [Papadimitriou & Yannakakis 1989]

- Width- w layered graph, edge lengths $l(e)$, first layer $\{s\}$, last layer $\{t\}$
- When a layer is reached, the next layer, the edges between the layers, and their lengths are revealed
- Goal: starting at s , reach t ; minimize total length of walk
- Deterministically: $O(w2^w) \cap \Omega(2^w) \cdot d_l(s, t)$ [Bur96,FFKRRRV91]
- Probabilistically: $O(w^{13}) \cap \Omega(w^2/\log^{1+\epsilon} w) \cdot d_l(s, t)$ [Ram93]
(for all $\epsilon > 0$)



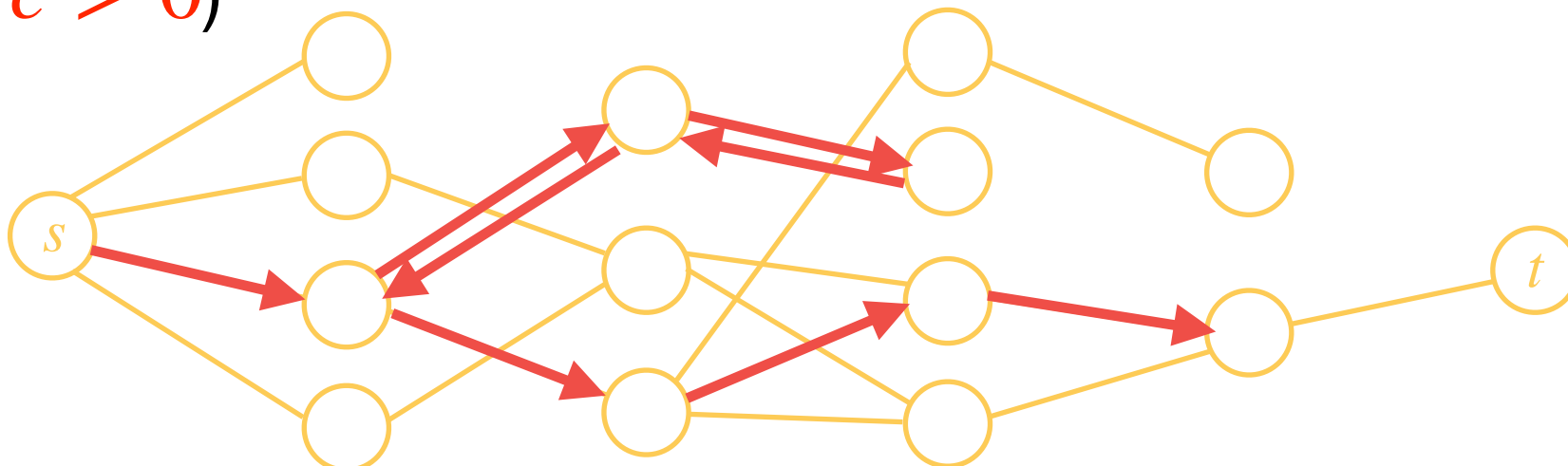
Layered graph traversal [Papadimitriou & Yannakakis 1989]

- Width- w layered graph, edge lengths $l(e)$, first layer $\{s\}$, last layer $\{t\}$
- When a layer is reached, the next layer, the edges between the layers, and their lengths are revealed
- Goal: starting at s , reach t ; minimize total length of walk
- Deterministically: $O(w2^w) \cap \Omega(2^w) \cdot d_l(s, t)$ [Bur96,FFKRRRV91]
- Probabilistically: $O(w^{13}) \cap \Omega(w^2/\log^{1+\epsilon} w) \cdot d_l(s, t)$ [Ram93]
(for all $\epsilon > 0$)



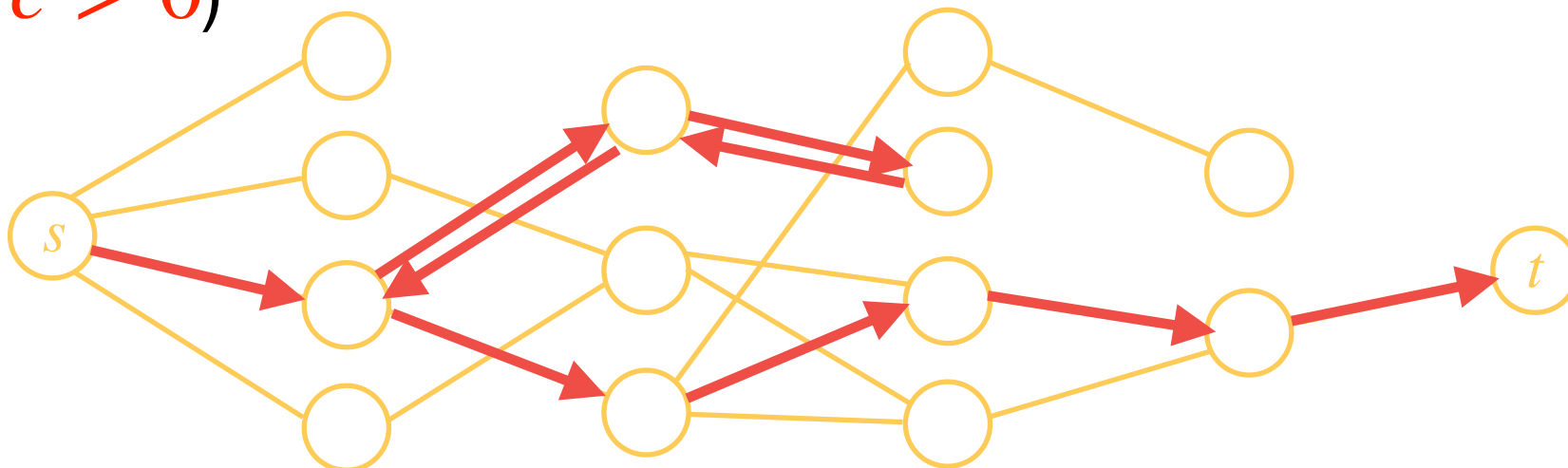
Layered graph traversal [Papadimitriou & Yannakakis 1989]

- Width- w layered graph, edge lengths $l(e)$, first layer $\{s\}$, last layer $\{t\}$
- When a layer is reached, the next layer, the edges between the layers, and their lengths are revealed
- Goal: starting at s , reach t ; minimize total length of walk
- Deterministically: $O(w2^w) \cap \Omega(2^w) \cdot d_l(s, t)$ [Bur96,FFKRRRV91]
- Probabilistically: $O(w^{13}) \cap \Omega(w^2/\log^{1+\epsilon} w) \cdot d_l(s, t)$ [Ram93]
(for all $\epsilon > 0$)



Layered graph traversal [Papadimitriou & Yannakakis 1989]

- Width- w layered graph, edge lengths $l(e)$, first layer $\{s\}$, last layer $\{t\}$
- When a layer is reached, the next layer, the edges between the layers, and their lengths are revealed
- Goal: starting at s , reach t ; minimize total length of walk
- Deterministically: $O(w2^w) \cap \Omega(2^w) \cdot d_l(s, t)$ [Bur96,FFKRRV91]
- Probabilistically: $O(w^{13}) \cap \Omega(w^2/\log^{1+\epsilon} w) \cdot d_l(s, t)$ [Ram93]
(for all $\epsilon > 0$)



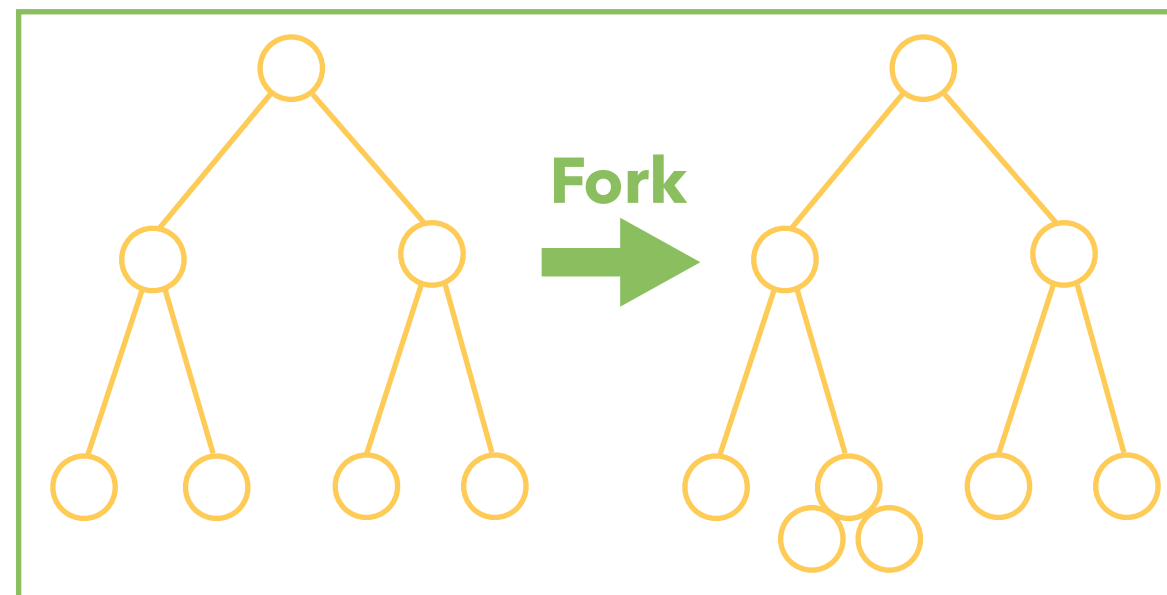
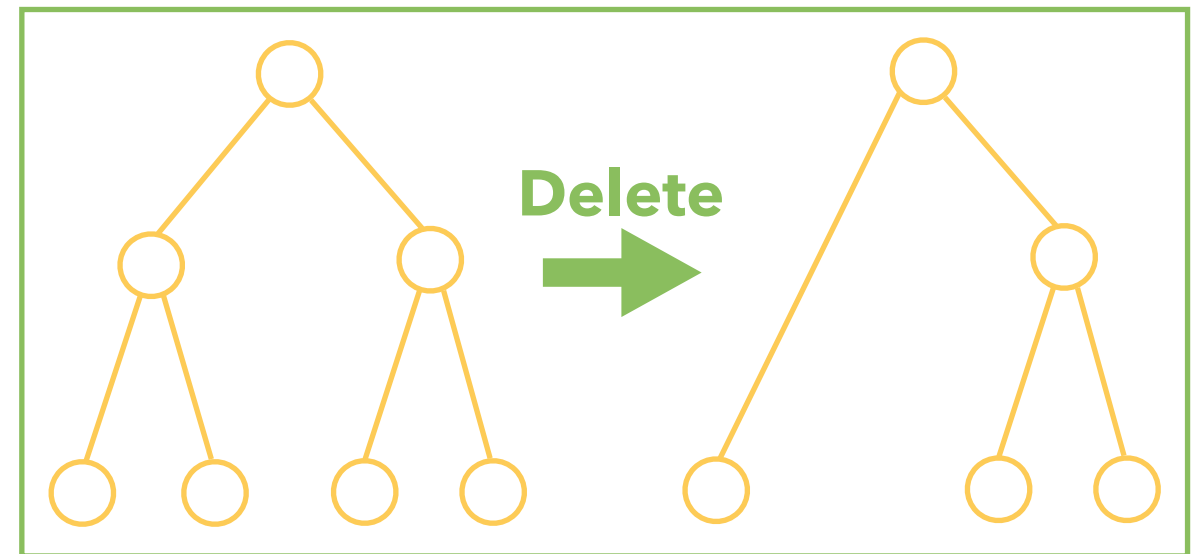
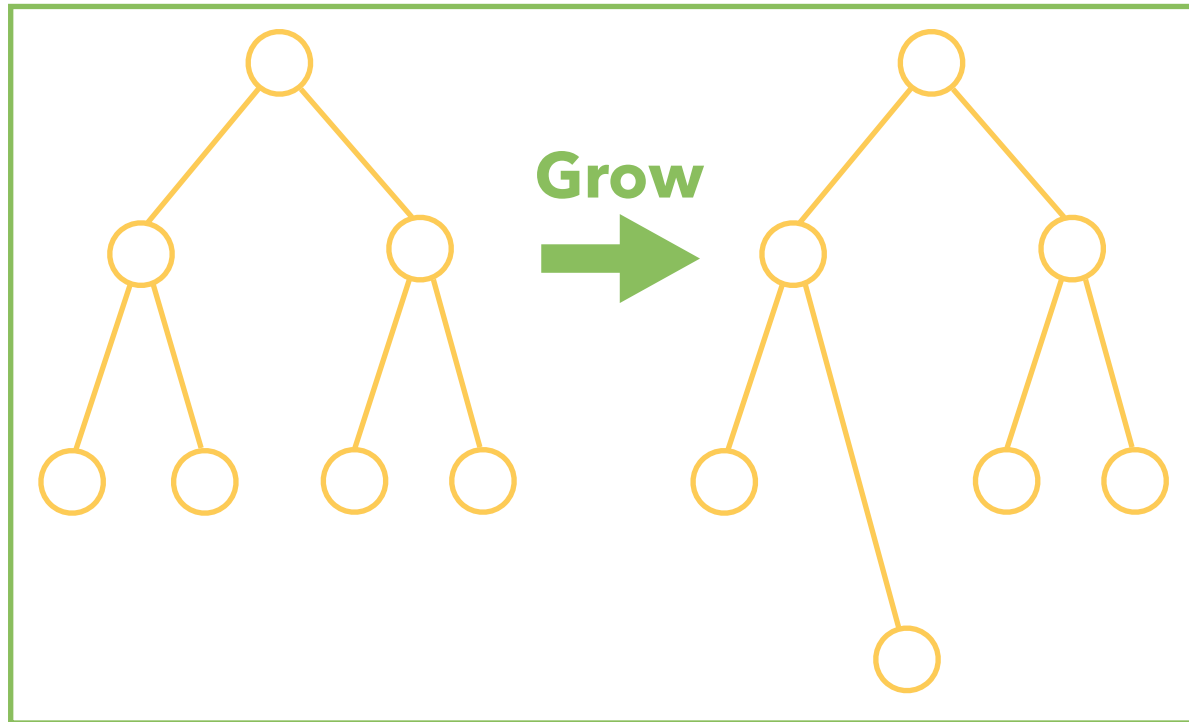
Small set chasing [Chrobak & Larmore 1991]

- a.k.a. metrical service systems
- Setting: metric space $\mathcal{M} = (X, d)$, initial state $\alpha_0 \in X$
- Pure ADV strategies: sequences $\sigma_1, \sigma_2, \sigma_3, \dots$ where $\forall i$,
 $\sigma_i \subset X, |\sigma_i| \leq w$
- Pure ALG strategies: sequences $\alpha_1, \alpha_2, \alpha_3, \dots$, where $\forall i, \alpha_i \in \sigma_i$
- Cost:
$$c_{\text{alg}}(\sigma, \alpha) = \sum_{i \geq 1} d(\alpha_{i-1}, \alpha_i)$$

Evolving tree game [Bubeck, Coester & R. 2022]

- Setting: initial edge-weighted tree T_0 rooted at r , with one other node c_r and an edge $\{r, c_r\}$ of length $l_{c_r} = 0$
- Pure ADV strategies: sequences $T_1, T_2, T_3, \dots$ where $\forall i, T_i$ is an edge-weighted tree rooted at r , derived from T_{i-1} by operations from:
 - Pick a leaf $x \neq r$ and increase the weight l_x of its edge to its parent
 - Delete a leaf $x \notin \{r, c_r\}$ and smooth the tree at its parent (may change c_r)
 - Generate ≥ 2 new nodes and attach them to a leaf x with edges of weight 0
- Pure ALG strategies: sequences $x_1, x_2, x_3, \dots$, where $\forall i, x_i$ is a leaf of T_i
- Cost: staying at x while l_x increases costs the increase, moving between leaves costs the current weight of the path connecting them

Evolving tree game (visualization)



Reductions

k -server in a $(k + 1)$ -point metric space

⋈ ⋈

[folklore]

set chasing in a $(k + 1)$ -point metric space

⋈

[folklore]

$(k + 1)$ -point metrical task systems

⋈

[R. Ph.D. thesis ?]

chasing $(k + 1)$ -point sets

⋈ ⋈

[FFKRRV91]

width- $(k + 1)$ layered graph traversal

⋈ ⋈

[FFKRRV91]

width- $(k + 1)$ layered tree traversal

⋈

[BCR22]

depth- $(k + 1)$ evolving (binary) tree game

Reductions

$\Theta(\log^2 k)$

k -server in a $(k + 1)$ -point metric space

$\bowtie$

[folklore]

set chasing in a $(k + 1)$ -point metric space

$\bowtie$

[folklore]

$(k + 1)$ -point metrical task systems

$\bowtie$

[R. Ph.D. thesis ?]

chasing $(k + 1)$ -point sets

$\bowtie$

[FFKRRV91]

width- $(k + 1)$ layered graph traversal

$\bowtie$

[FFKRRV91]

width- $(k + 1)$ layered tree traversal

$\bowtie$

[BCR22]

depth- $(k + 1)$ evolving (binary) tree game

Reductions

$\Theta(\log^2 k)$

k -server in a $(k + 1)$ -point metric space

$\bowtie$

[folklore]

set chasing in a $(k + 1)$ -point metric space

$\bowtie$

[folklore]

$(k + 1)$ -point metrical task systems

$\bowtie$

[R. Ph.D. thesis ?]

chasing $(k + 1)$ -point sets

$\bowtie$

[FFKRRV91]

width- $(k + 1)$ layered graph traversal

$\bowtie$

[FFKRRV91]

width- $(k + 1)$ layered tree traversal

$\bowtie$

[BCR22]

depth- $(k + 1)$ evolving (binary) tree game

$\Theta(k^2)$

Our results

- For all n there exist n -point metric spaces where the rand. c.r. for chasing $(n - 1)$ -point sets is in $\Omega(\log^2 n)$
 $\Rightarrow$ lower bounds for k -server ($k = n - 1$), n -point MTS, other problems

Matches the $O(\log^2 n)$ universal upper bound for n -point MTS

First improvement of existential l.b. of [BLS87]

- For all n -point metric spaces: rand. c.r. for set chasing is in $\Omega(\log n)$
 $\Rightarrow$ universal lower bounds (matching existential upper bounds)

- The rand. c.r. for width- n LGT is in $\Omega(n^2)$
 $\Rightarrow$ lower bounds for chasing n -point sets, depth- n ETG

Matches the $O(n^2)$ upper bound for the depth- n evolving tree game

Broader context: online selection

- Sequence of sets; select a point in each set; some measure of distance:
 - Convex bodies in ℓ_p^n [Sel20,AGGT20] n
(tight lower bound for $p = \infty$ [BKLLS20 ?])
 - Subsets of an n -point metric (a.k.a. MTS) [BCLL19,BCR23] $\Theta(\log^2 n)$
(faces of the n -dimensional simplex under a transportation norm)
 - n -point sets of an infinite metric (a.k.a. LGT) [BCR22,BCR23] $\Theta(n^2)$

[BLS87] lower bound

- Uniform n -point metric space
- Use Yao's minimax
- Random sequence: i.i.d. requests u.a.r. among all points
chosen point has cost 1 , all other points have cost 0
- $E[c_{\text{alg}}] = \text{\#requests} / n$
- The adversary can move to the last point to be requested, restarting each time this position is requested
- $E[c_{\text{base}}] = \text{\#requests} / \Omega(n \log n)$ (coupon collector)

An $\Omega((\log n / \log \log n)^2)$ lower bound

- Metric spaces $\mathcal{M}_w = (X_w, d_w)$, for all $w \in \mathbb{N} \cup \{0\}$
- d_w is the shortest path metric of an unweighed graph
- Select two terminals $s_w, t_w \in X_w$
- $n = |X_w| = \exp(w \log w)$
- We'll show a lower bound of $\Omega(w^2) = \Omega((\log n / \log \log n)^2)$
- Choose natural numbers $m_0 \leq m_1 \leq m_2 \leq \dots$ TBD

Some intuition

The Road Not Taken BY ROBERT FROST

Two roads diverged in a yellow wood,
And sorry I could not travel both
And be one traveler, long I stood
And looked down one as far as I could
To where it bent in the undergrowth;

Then took the other, as just as fair,
And having perhaps the better claim,
Because it was grassy and wanted wear;
Though as for that the passing there
Had worn them really about the same,

And both that morning equally lay
In leaves no step had trodden black.
Oh, I kept the first for another day!
Yet knowing how way leads on to way,
I doubted if I should ever come back.

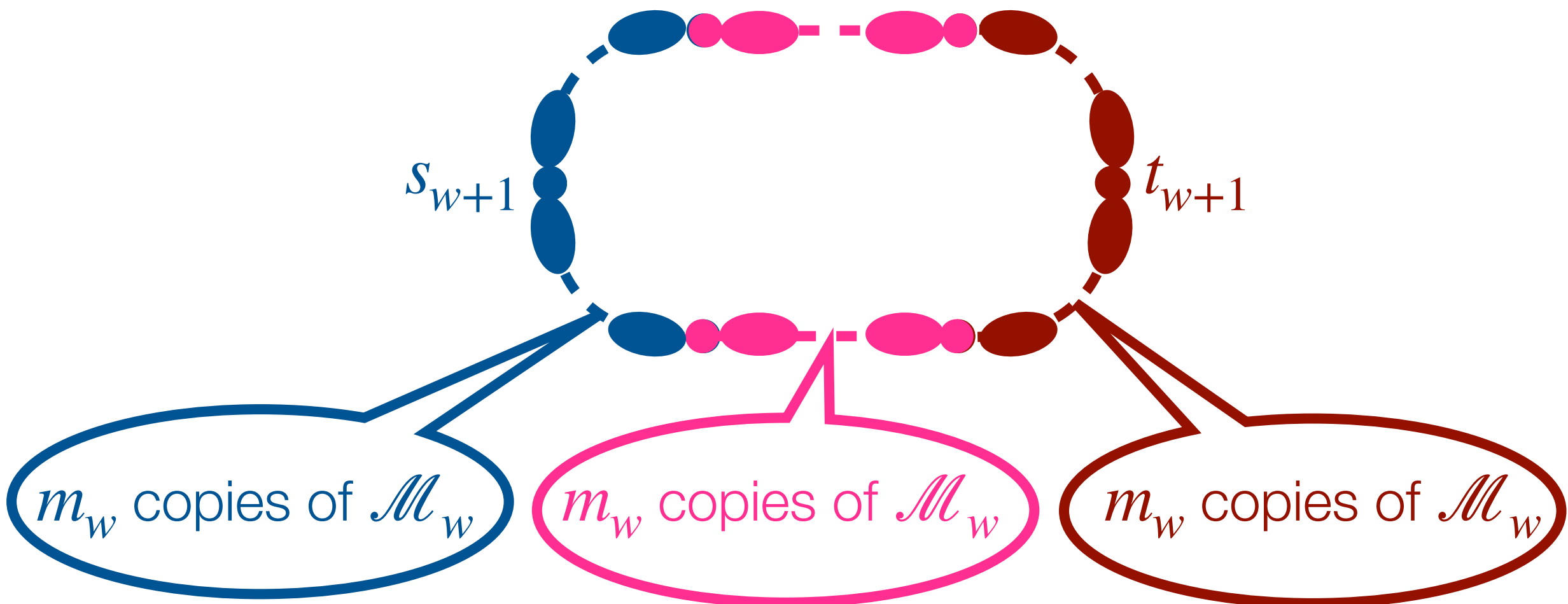
I shall be telling this with a sigh
Somewhere ages and ages hence:
Two roads diverged in a wood, and I—
I took the one less traveled by,
And that has made all the difference.

The bad metric space

- Base case: $\mathcal{M}_0$ is an edge ($X_0 = \{s_0, t_0\}$ and $d_0(s_0, t_0) = 1$)

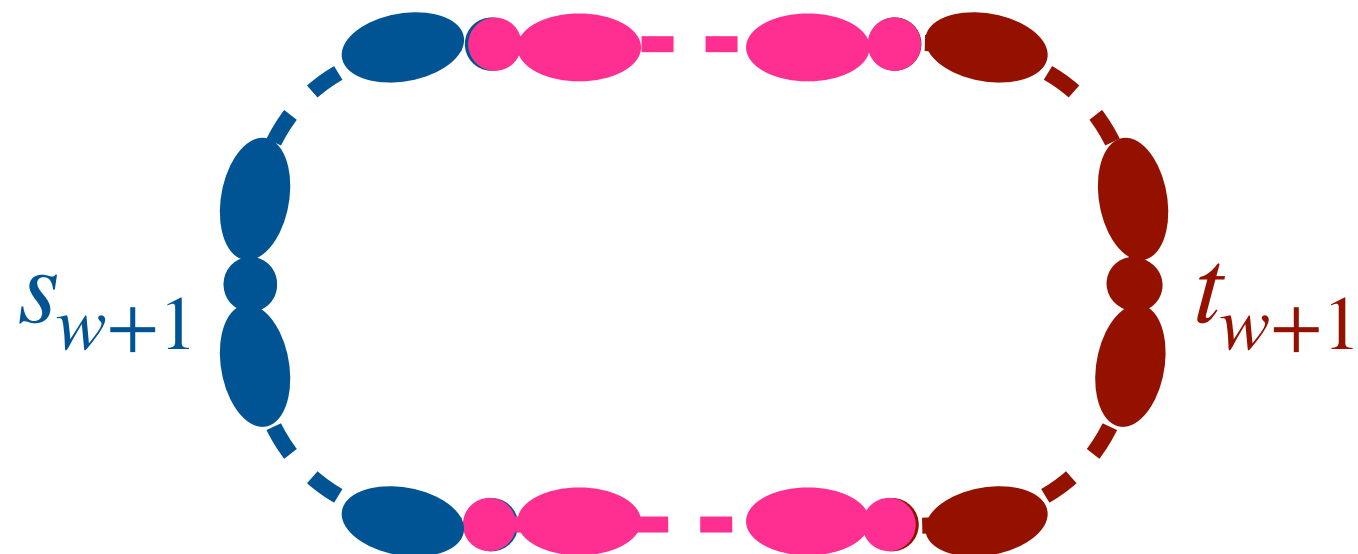


- Inductive step: $\mathcal{M}_{w+1}$ is a cycle of $6m_w$ copies of $\mathcal{M}_w$



The bad sequence $\rho(\mathcal{M}_{w+1})$

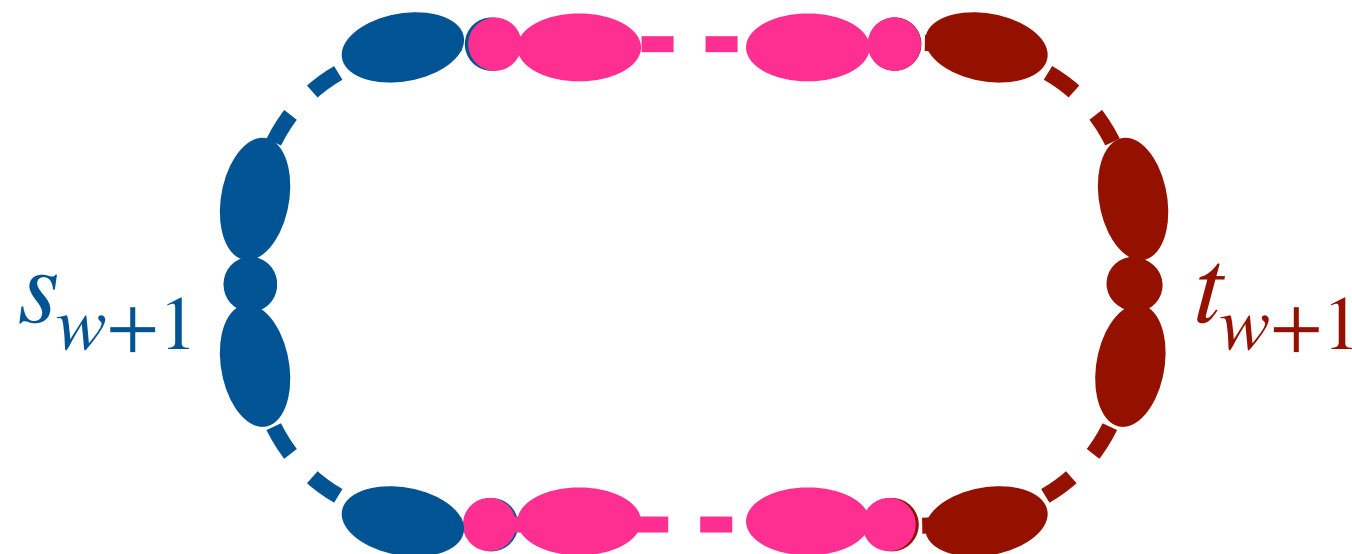
- Top path: $\mathcal{M}_{w,t,1}, \mathcal{M}_{w,t,2}, \dots, \mathcal{M}_{w,t,3m_w}$
bottom path: $\mathcal{M}_{w,b,1}, \mathcal{M}_{w,b,2}, \dots, \mathcal{M}_{w,b,3m_w}$
- Stage 1: $\rho(\mathcal{M}_{w,t,1}) \oplus \rho(\mathcal{M}_{w,b,1}), \dots, \rho(\mathcal{M}_{w,t,m_w}) \oplus \rho(\mathcal{M}_{w,b,m_w})$
- Stage 2: for m_w steps, for $p \in_{i.\&u.a.r} \{t, b\}$, use $\rho(\mathcal{M}_{w,p,next(p)}) \oplus s_{w,\bar{p},next(\bar{p})}$
- Stage 3: (w.l.o.g t is the road less traveled by) $\rho(\mathcal{M}_{w,t,next(t)}), \dots, \rho(\mathcal{M}_{w,t,3m_w})$



The bad sequence $\rho(\mathcal{M}_{w+1})$

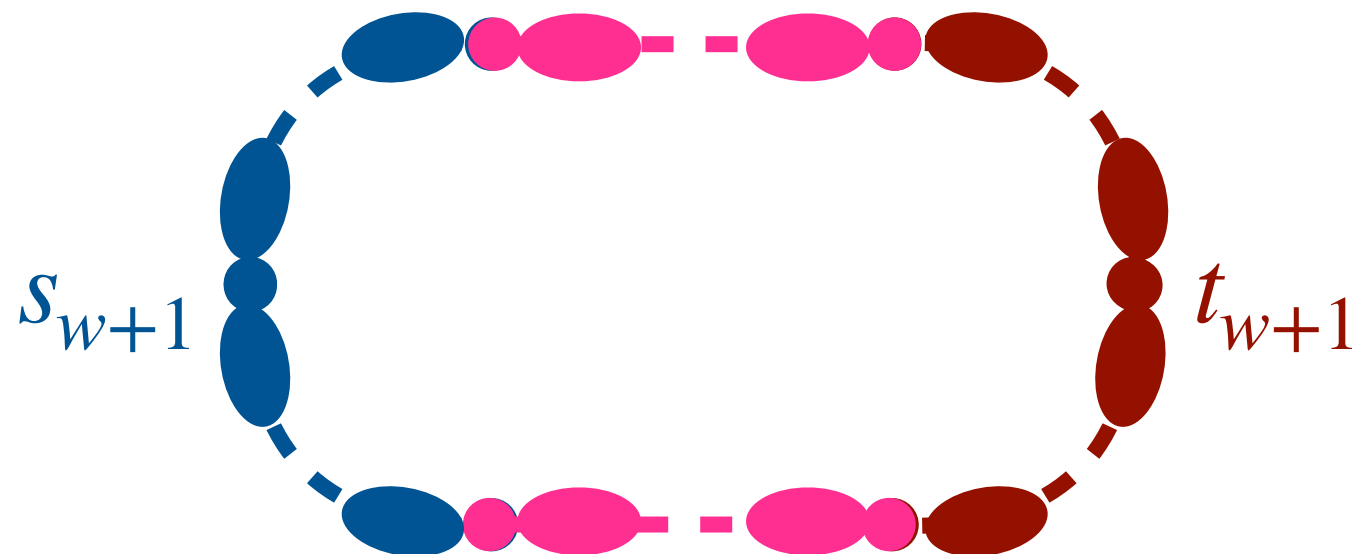
- Top path: $\mathcal{M}_{w,t,1}, \mathcal{M}_{w,t,2}, \dots, \mathcal{M}_{w,t,3m_w}$
bottom path: $\mathcal{M}_{w,b,1}, \mathcal{M}_{w,b,2}, \dots, \mathcal{M}_{w,b,3m_w}$
- Stage 1: $\rho(\mathcal{M}_{w,t,1}) \oplus \rho(\mathcal{M}_{w,b,1}), \dots, \rho(\mathcal{M}_{w,t,m_w}) \oplus \rho(\mathcal{M}_{w,b,m_w})$
- Stage 2: for m_w steps, for $p \in_{i.\&u.a.r} \{t, b\}$, use $\rho(\mathcal{M}_{w,p,next(p)}) \oplus s_{w,\bar{p},next(\bar{p})}$
- Stage 3: (w.l.o.g t is the road less traveled by) $\rho(\mathcal{M}_{w,t,next(t)}), \dots, \rho(\mathcal{M}_{w,t,3m_w})$

The next $\mathcal{M}_w$ copy
on path p

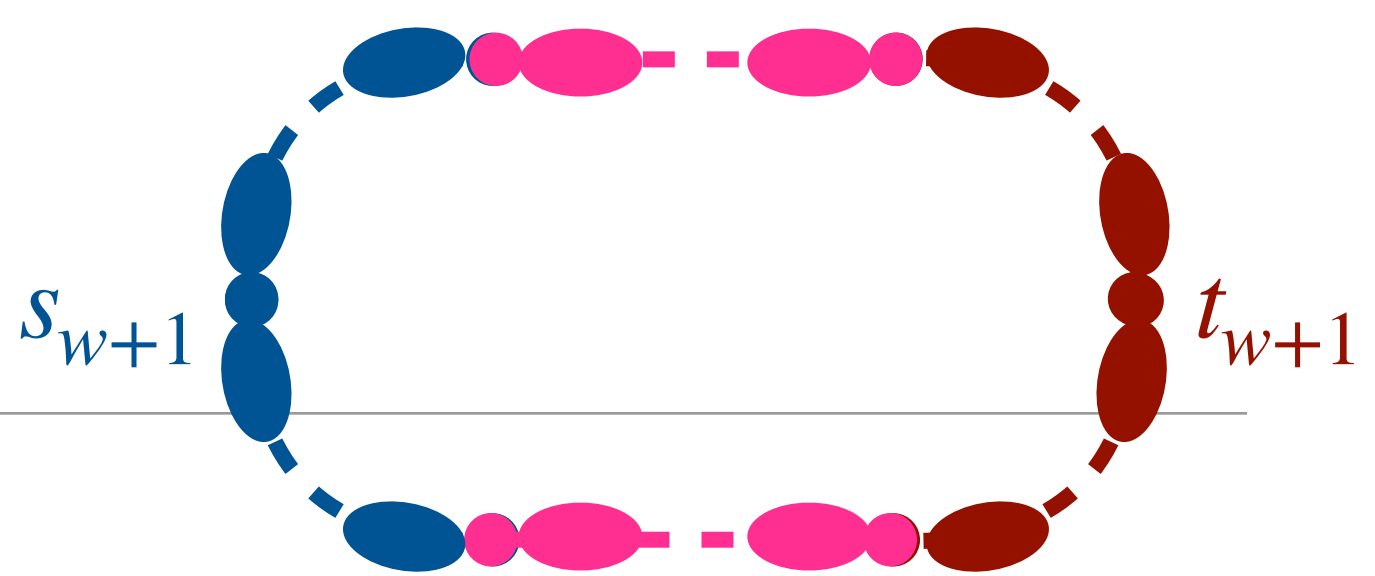


The bad sequence $\rho(\mathcal{M}_{w+1})$

- Top path: $\mathcal{M}_{w,t,1}, \mathcal{M}_{w,t,2}, \dots, \mathcal{M}_{w,t,3m_w}$
bottom path: $\mathcal{M}_{w,b,1}, \mathcal{M}_{w,b,2}, \dots, \mathcal{M}_{w,b,3m_w}$
- Stage 1: $\rho(\mathcal{M}_{w,t,1}) \oplus \rho(\mathcal{M}_{w,b,1}), \dots, \rho(\mathcal{M}_{w,t,m_w}) \oplus \rho(\mathcal{M}_{w,b,m_w})$
- Stage 2: for m_w steps, for $p \in_{i.\&u.a.r} \{t, b\}$, use $\rho(\mathcal{M}_{w,p,next(p)}) \oplus s_{w,\bar{p},next(\bar{p})}$
- Stage 3: (w.l.o.g t is the road less traveled by) $\rho(\mathcal{M}_{w,t,next(t)}), \dots, \rho(\mathcal{M}_{w,t,3m_w})$



Analysis



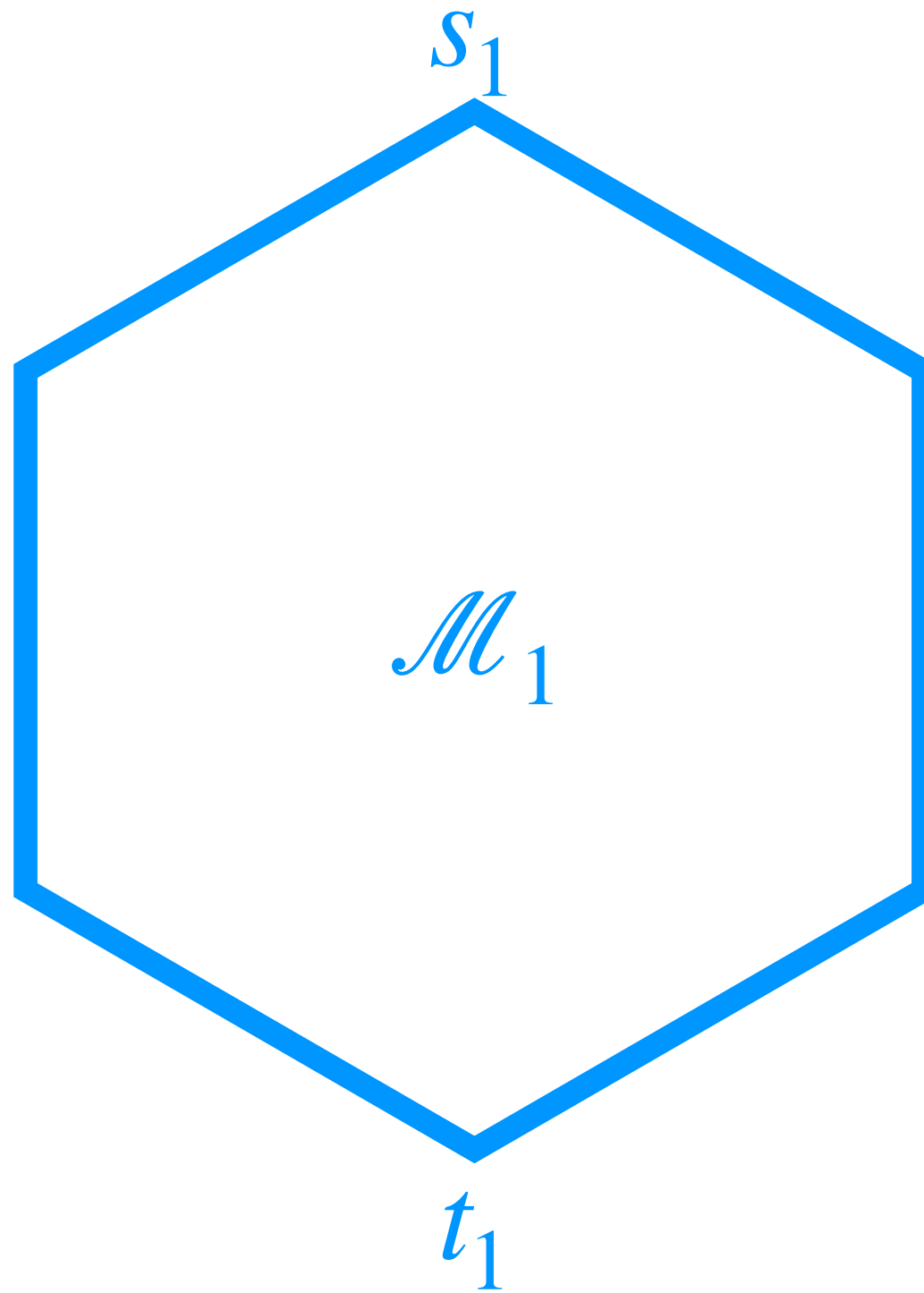
- Let $C_w :=$ lower bound for $\mathcal{M}_w$
- By induction $c_{\text{base}}(\rho(\mathcal{M}_w)) = d_w(s_w, t_w)$
- The anti-server starts at s_w and finishes at t_w
- Stage 1 in $\mathcal{M}_{w+1}$ costs at least $m_w C_w d_w(s_w, t_w)$
- Stage 2 costs at least $(m_w/2) C_w d_w(s_w, t_w)$
(requires proof that switching paths doesn't help)
- At the end of stage 2, w.h.p. $\text{next}(t) = (3m_w/2) - \Theta(\sqrt{m_w})$,
so stage 3 costs at least $(3m_w/2 + \Theta(\sqrt{m_w})) C_w d_w(s_w, t_w)$
- For $m_w = C_w$ we get $C_{w+1} = C_w + \Theta(\sqrt{C_w}) \implies C_w = \Omega(w^2)$

Comments

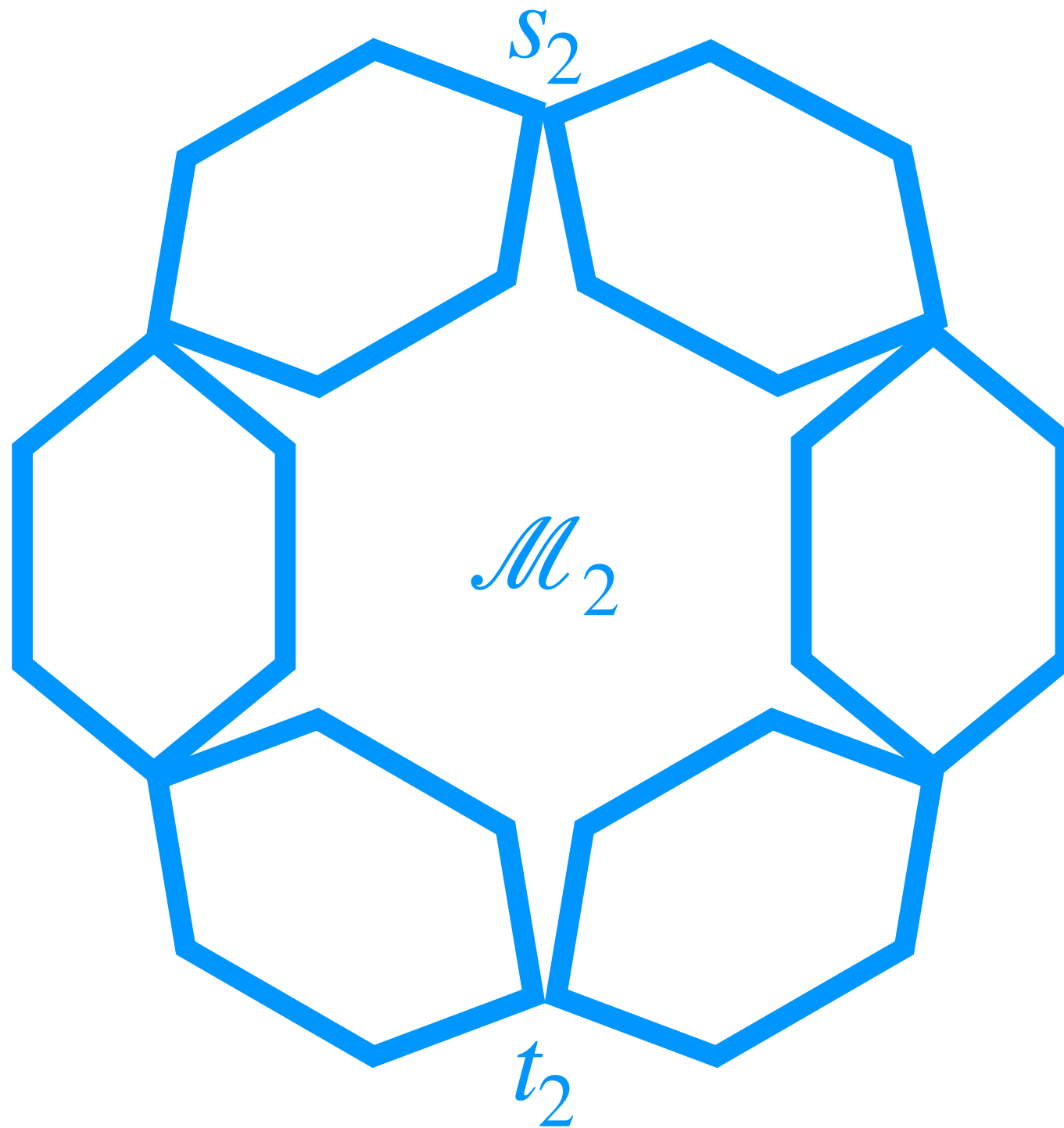
- The bad sequence can be mirrored from t_w to s_w ; alternating directions, we can repeat many times.
- To get $\Omega(\log^2 n)$: “shrink” $\mathcal{M}_{w+1}$ to a cycle of just 6 copies of $\mathcal{M}_w$.
- Stage 2 requires “breaking” the bad sequences for the middle copies in the top and bottom paths into chunks.
- The choices to “hit” top or bottom are no longer independent.
- The analysis requires a lower bound on the deviation of a martingale. We use a theorem of Ibragimov (1963).
- For the LGT lower bound, we need to control the size of each request.

The $\Omega(\log^2 n)$ metric

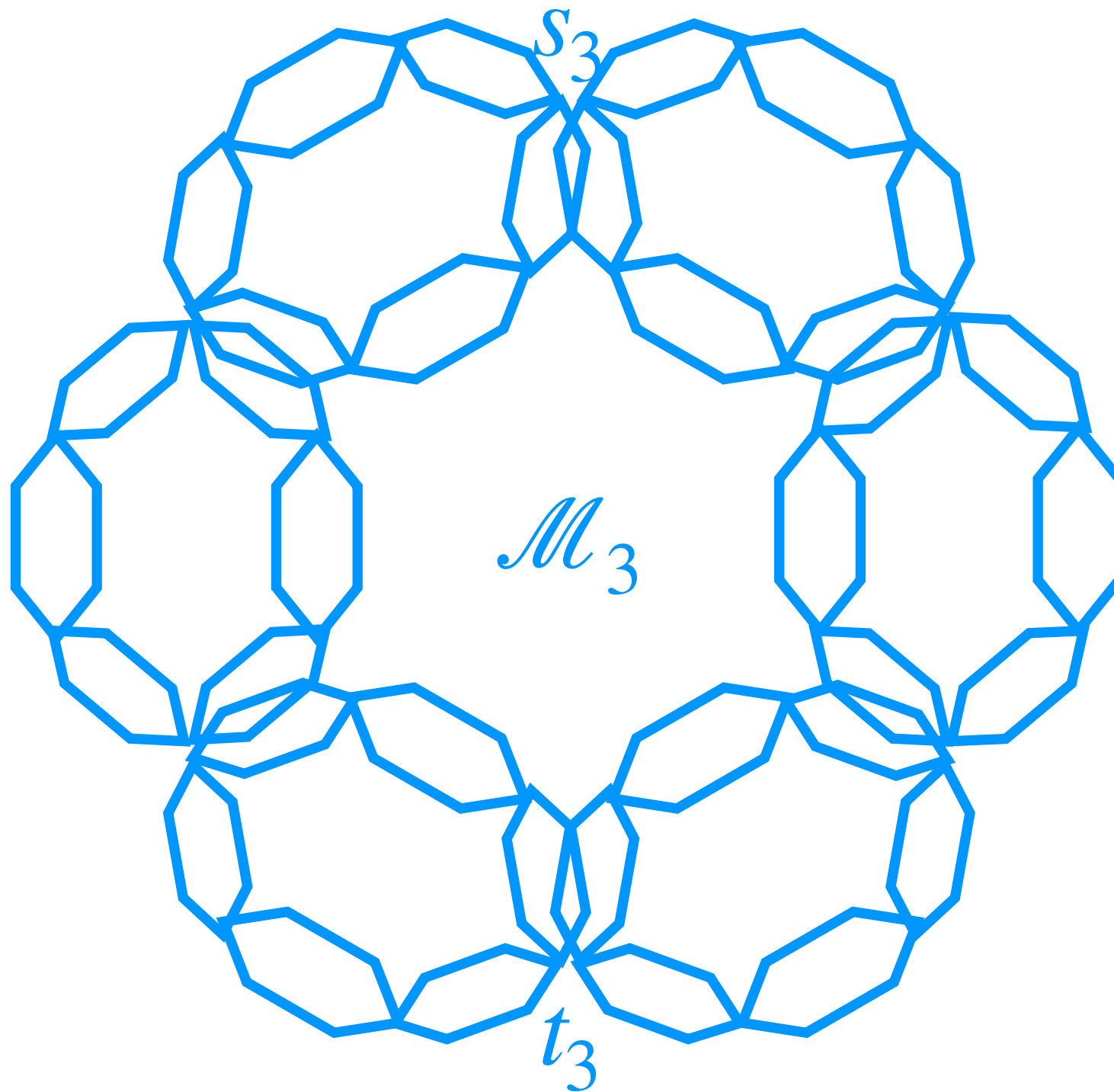
The $\Omega(\log^2 n)$ metric



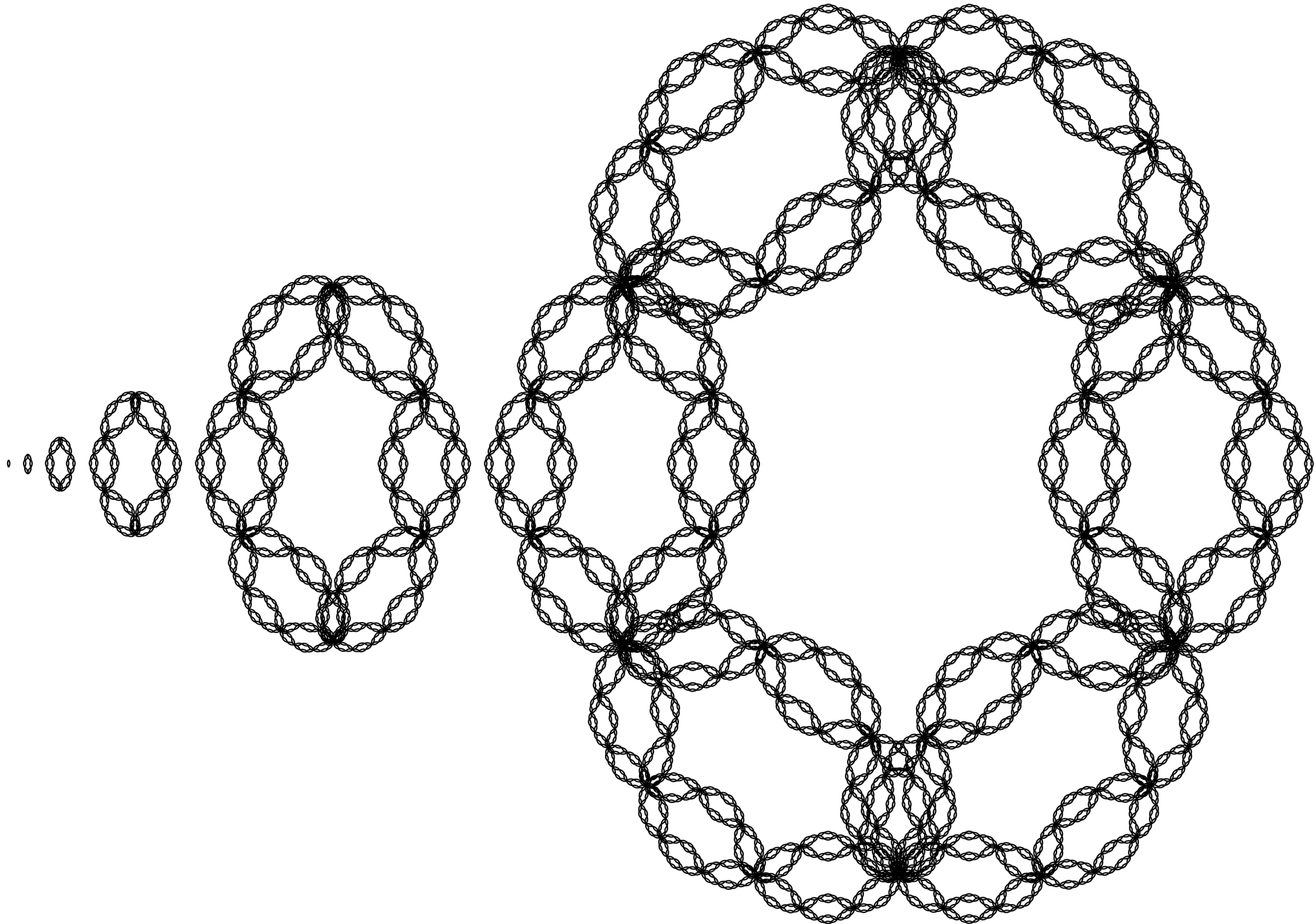
The $\Omega(\log^2 n)$ metric



The $\Omega(\log^2 n)$ metric



The $\Omega(\log^2 n)$ metric



Some notes on the ETG upper bound

- Setting: initial edge-weighted tree T_0 rooted at r , with one other node c_r and an edge $\{r, c_r\}$ of length $l_{c_r} = 0$
- The adversary (MAX player) can:
 - Pick a leaf $v \neq r$ and increase the weight l_v of its edge to its parent
 - Delete a leaf $\notin \{r, c_r\}$ and smooth the tree at its parent (may change c_r)
 - Generate ≥ 2 new nodes and attach them to a leaf with edges of weight 0
- The online algorithm must occupy a leaf at all times
- Cost:
 - staying at a leaf v while l_v increases costs the increase
 - moving to a leaf costs the current weight of the traversed path

Motivation

- Mirror descent: strongly convex regularizer Φ , control function f ,
dynamic: $\nabla^2 \Phi(x) \frac{dx}{dt} \in f(x) - N(x)$ [NY79, BCLLM18]
 $N(x)$ is the normal cone at x of the feasible region
- The Bregman divergence $D_\Phi(y \parallel x) \triangleq \Phi(y) - \Phi(x) - \langle \nabla \Phi(x), y - x \rangle$
is a natural potential function: $\frac{dD_\Phi(y \parallel x)}{dt} \leq \langle -f, y - x \rangle$
 $-f$ is the inflicted loss, y is MAX (a.k.a. adversary, or KAOS*),
 x is the algorithm (a.k.a. the good guys, or CONTROL) _ _

* Has agent 86 switched sides?! 😓



Our setting

Entropic regularizer is a natural choice: $\Phi(x) = \sum_{v \neq r} l_v(x_v + \delta_v) \log(x_v + \delta_v)$

The natural control is $f(x) = -\frac{dl}{dt}$

This gives $D_\Phi(y \parallel x) = \sum_{u \neq r} l_u \left((y_u + \delta_u) \log \frac{y_u + \delta_u}{x_u + \delta_u} + x_u - y_u \right)$

So, $\frac{dD_\Phi(y \parallel x)}{dt} = y_v \cdot O(\log(1/\delta_v)) + \delta_v \cdot O(\log(1/\delta_v)) + x_v$

The $+x_v$ term can be dealt with by doubling the rate (multiply f by 2).

The $\delta_v \cdot O(\log(1/\delta_v))$ term can't be controlled, so we remove it from Φ

This takes care of the “service cost”

Movement cost

It turns out we must slow down when x_v is small.

So, the revised control function is $f_v = -\frac{2x_v}{x_v + \delta_v} \cdot \frac{dl_v}{dt}$

Now the weighted depth potential $-\Psi(x) = -\sum_{u \neq r} h_u l_u x_u$ relates the movement cost to the loss under f (a.k.a. service cost)

But Ψ doesn't behave well under topology changes, hence instead of l_u we use $\tilde{l}_u = \frac{2k-1}{2k-h_u} (l_u + \epsilon 2^{-j_u})$ ($j_u =$ step in which u was created)

Concluding remarks

- Resolves MTS, LGT/MSS/ETG;
surprise: some metrics are more equal than others!
- MTS vs ETG — upper bounds use the mirror descent schema:
 - MTS on a hierarchically separated n -leaf tree — c.r. $\Theta(\log n)$:
 Use conditional entropy [CL20] $\Phi(x) = \sum_{u \neq r} \frac{l_u}{\eta_u} (x_u + \delta_u x_{p_u}) \log \left(\frac{x_u}{x_{p_u}} + \delta_u \right)$
 c.r. $\log \# \text{leaves}$ ($\times$ distortion of embedding into HSTs, unavoidable!)
 - Evolving depth- k tree game — c.r. $\Theta(k^2)$:
 Use uncond. entropy $\Phi(x) = \frac{1}{\eta} \sum_{u \neq r} \tilde{l}_u x_u \log(x_u + \delta_u)$ and
 $-\Psi(x) = - \sum_{u \neq r} h_u \tilde{l}_u x_u$ c.r. $\log \# \text{leaves} \times \text{tree depth}$
- MTS: What's the truth for an n -path? Parameterization of all metrics?
 Is it always $O(\log n)$ $\times$ distortion of embedding into HSTs?